Report

Of

GenAI Powered Data Analytics

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 25SCS1003005399

Name of Student: SYED ARSHADUL HAQUE

Batch: 2025-29

CANDIDATE'S DECLARATION

I, Syed Arshadul do hereby declare that the internship report titled **"**GenAI Powered Data Analytics**"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

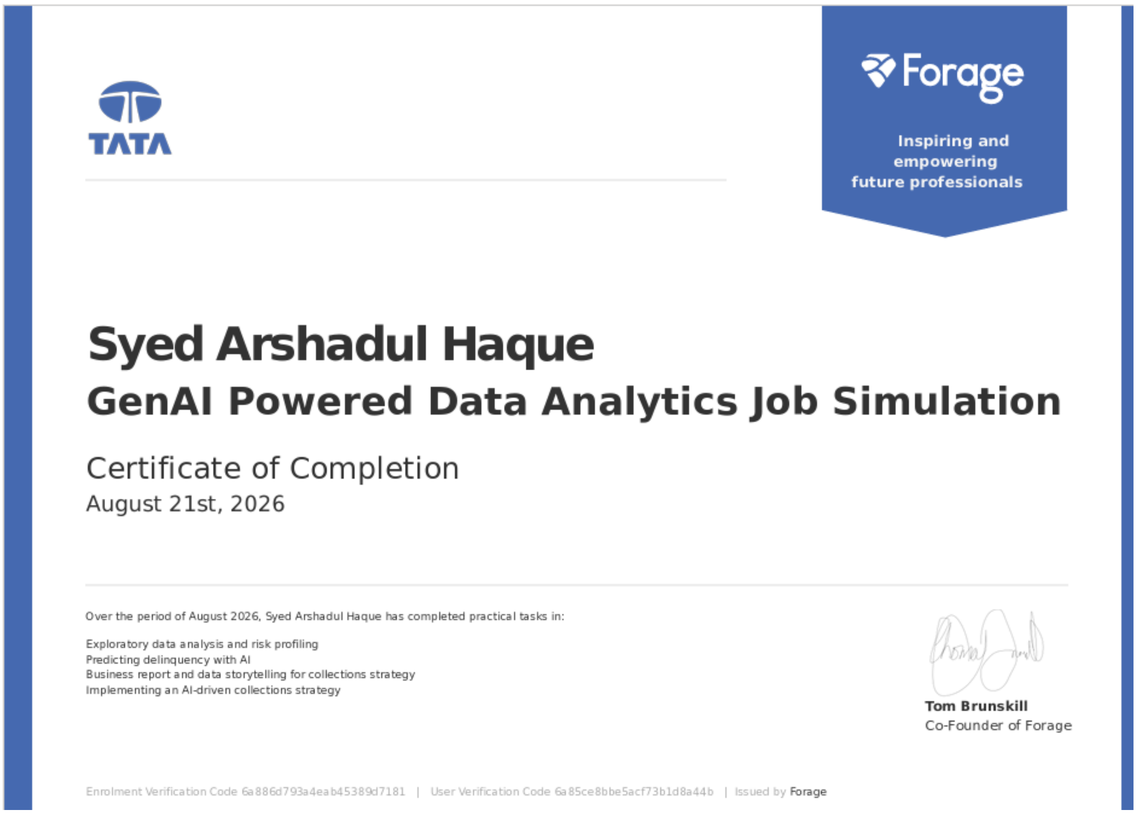
Signature
Student Name: Syed Arshadul Haque
Roll No.: 25SCS1003005399

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.
I express my gratitude to my parents for their blessings.

Signature
Student Name: Syed Arshadul Haque
Roll No.: 25SCS1003005399

Completion Certificate


TATA

Forage

Inspiring and empowering future professionals

Syed Arshadul Haque

GenAI Powered Data Analytics Job Simulation

Certificate of Completion

August 21st, 2026

Over the period of August 2026, Syed Arshadul Haque has completed practical tasks in:

Exploratory data analysis and risk profiling
Predicting delinquency with AI
Business report and data storytelling for collections strategy
Implementing an AI-driven collections strategy

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a886d793a4eab45389d7181 | User Verification Code 6a85ce8bbe5acf73b1d8a44b | Issued by Forage

4.1 Introduction

Artificial Intelligence (AI) and Generative AI (GenAI) are revolutionizing the way organizations approach data analytics. Traditional analytics methods often rely on static rules and predefined models, which can be limiting when dealing with complex, dynamic, or unstructured datasets. In contrast, AI driven systems can learn from data, adapt to new information, and generate insights that are both accurate and actionable. GenAI further enhances this process by creating synthetic data, simulating scenarios, and generating predictions that improve decision making.

The GenAI Powered Data Analytics Job Simulation **at** Tata Forage was designed to provide students with practical exposure to these cutting edge technologies. Conducted in August 2026, the program offered a structured curriculum that combined theoretical learning with hands on tasks. Participants were introduced to industry level datasets, modern AI frameworks, and workflows that mirror real world applications.

The internship emphasized the following key areas:

- Data Preprocessing**:** Techniques for cleaning, organizing, and preparing raw datasets to ensure reliability.
- Machine Learning Algorithms**:** Application of regression, classification, and clustering methods to solve practical problems.
- Generative AI Models: Use of advanced frameworks to enhance analytics, generates synthetic data, and improves predictive accuracy.
- Evaluation Metrics: Understanding performance measures such as accuracy, precision, recall, and F1-score.
- Ethical AI Practices**:** Ensuring fairness, transparency, and accountability in AI driven decision making.

By engaging in this program, I was able to bridge the gap between classroom learning and industry practice. The internship not only strengthened my technical skills in Python, data handling, and model training but also enhanced my professional competencies such as documentation, systematic working, and time management

4.2 Organization Profile

The internship was undertaken at **Tata Forage**, a global learning platform that provides industry recognized job simulations. These programs are designed to replicate real workplace challenges and allow students to gain practical exposure while still pursuing their academic studies.

The **GenAI Powered Data Analytics Job Simulation** was conducted in **August 2026**. It was specifically designed to help students understand how Artificial Intelligence and Generative AI can be applied in the field of data analytics. The program combined theoretical modules with practical assignments, ensuring that participants not only learned the concepts but also applied them to solve realistic problems.

Key Details of the Organization and Program:

- **Organization Name:** Tata Forage
- **Program Name:** GenAI Powered Data Analytics
- **Role:** Data Analytics Intern
- **Date of completion :** 21 August 2026
- **Field:** Artificial Intelligence & Data Analytics

Through this program, I was introduced to curated datasets, modern AI frameworks, and workflows that mirror industry practices. The structured curriculum helped me connect classroom knowledge with professional applications, strengthen my programming skills, and gain confidence in applying generative AI techniques to data driven problem solving.

4.3 Problem Statement

In the modern digital era, organizations generate and collect massive amounts of data every day. This data comes from diverse sources such as customer transactions, social media interactions, sensor readings, and enterprise systems. While the availability of such large datasets creates opportunities for deeper insights, it also presents significant challenges in terms of storage, processing, and analysis.

Traditional data analytics methods often rely on static models and rule-based approaches. These techniques can be effective for small, structured datasets but struggle when applied to large, complex, or unstructured data. As a result, organizations face difficulties in extracting meaningful patterns, predicting future trends, and making informed decisions.

The **GenAI Powered Data Analytics Job Simulation** was designed to address these challenges by introducing students to advanced techniques in Artificial Intelligence and Generative AI. The central problem identified during the internship was **how to transform raw, complex datasets into actionable intelligence using AI driven methods**.

Key Issues

1. **Data Quality Problems:** Handling missing values, duplicates, and inconsistencies in datasets.
2. **Complexity of Datasets:** Managing large volumes of structured and unstructured data.
3. **Algorithm Selection:** Choosing suitable machine learning and generative models for different types of problems.
4. **Model Training and Testing:** Ensuring accuracy while avoiding over fitting or under fitting.
5. **Interpretability of Results:** Making AI generated insights understandable and useful for decision makers.
6. **Bridging Theory and Practice:** Applying classroom knowledge to industry level problems in a systematic way.

4.4 Project Objectives

The main objective of the internship project was to gain practical exposure to the applications of Artificial Intelligence and Generative AI in the field of data analytics. While classroom learning provides theoretical knowledge, this program was designed to help students apply those concepts to real datasets and industry-level problems.

The specific objectives of the project were:

To understand the fundamentals of Artificial Intelligence and Generative AI and how these technologies are transforming data analytics in modern organizations.

To strengthen programming skills in Python and learn how to use essential libraries such as NumPy, Pandas, Scikit-learn, and TensorFlow for data handling and model development.

To gain practical knowledge of data preprocessing techniques including cleaning, normalization, and handling missing values, which are critical for ensuring the accuracy of analytics.

To apply machine learning algorithms and generative models for tasks such as classification, regression, clustering, and predictive analytics.

To evaluate model performance using standard metrics like accuracy, precision, recall, and F1-score, and to understand how to improve results through systematic testing.

To develop professional skills such as documentation, systematic working, and time management, which are essential for success in industry projects.

To bridge the gap between academic learning and industry practice by working on curated datasets and workflows that mirror real organizational challenges.

To build a strong foundation for future projects in Artificial Intelligence and Data Science by gaining confidence in applying theoretical concepts to practical scenarios

4.5 Scope of the Project

The scope of the project undertaken during the internship was wide and covered both technical learning and professional development. The GenAI Powered Data Analytics Job Simulation was designed to provide exposure to the complete cycle of data analytics, beginning from problem identification and extending to the delivery of meaningful insights using Artificial Intelligence and Generative AI techniques.

The project focused on understanding the fundamentals of AI and GenAI and their role in modern data analytics. It included learning the basics of machine learning, such as supervised and unsupervised techniques, and applying them to curated datasets. A major part of the scope was dedicated to data preprocessing, which involved cleaning, normalization, and handling missing values to ensure the reliability of results.

Another important aspect was the application of Python programming and libraries like NumPy, Pandas, Scikit learn, and TensorFlow to build and test models. Generative AI models were explored to enhance analytics by generating synthetic data and improving predictive accuracy. The scope also extended to evaluating model performance using standard metrics and identifying methods to improve accuracy and reliability.

Beyond technical learning, the project emphasized professional skills such as systematic working, documentation, and time management. By working on tasks that mirrored real organizational challenges, the internship helped bridge the gap between academic learning and industry practice. The overall scope ensured that the internship was not limited to theoretical knowledge but provided a holistic experience that prepared me for advanced studies and future career opportunities in Artificial Intelligence and Data Science.

4.6 Technologies and Tools Used

The successful completion of the internship project required the use of several technologies and tools that supported data handling, model building, and evaluation. The primary programming language used was Python, which is widely recognized for its simplicity and extensive support in data science and artificial intelligence. Python provided access to multiple libraries that simplified tasks such as preprocessing, visualization, and implementation of machine learning algorithms.

Some of the major technologies and tools applied during the project were:

- **Python Programming Language** – served as the foundation for all coding tasks, offering flexibility and a wide range of libraries for AI and data analytics.
- **NumPy and Pandas** – used for numerical computing and data manipulation, enabling efficient handling of large datasets.
- **Matplotlib and Seaborn** – applied for data visualization, allowing graphical representation of trends, distributions, and model results.
- **Scikit learn** – implemented for machine learning tasks such as regression, classification, and clustering.
- **TensorFlow and Keras** – explored for deep learning and generative AI models, providing advanced capabilities for predictive analytics.
- **Jupyter Notebook** – used as the interactive environment for coding, analysis, and documentation of results.
- **GitHub** – utilized for version control and maintaining organized project files.

Each of these tools played a specific role in different stages of the project. For example, Pandas was essential for cleaning and preparing datasets, while Scikit learn provided ready-to-use algorithms for model development. TensorFlow and Keras allowed experimentation with generative AI models, which enhanced the scope of analytics by generating synthetic data and improving predictions. Visualization libraries like Matplotlib and Seaborn ensured that results were presented in a clear and understandable manner

4.7 System Architecture

The system architecture designed for the internship project represented the complete workflow of data analytics using Artificial Intelligence and Generative AI techniques. It provided a structured approach to problem solving and ensured that each stage contributed to the accuracy and reliability of the final results.

The process began with the input of datasets, which were collected and prepared for analysis. This was followed by data preprocessing, where tasks such as cleaning, normalization, and handling missing values were performed to improve the quality of data. Once the data was ready, exploratory data analysis was carried out to identify patterns, correlations, and trends that could guide the selection of suitable models.

Based on the requirements of the problem, appropriate machine learning algorithms or generative AI models were chosen. These models were trained using the prepared datasets and tested to validate their performance. The evaluation stage involved measuring accuracy, precision, recall, and other metrics to assess the effectiveness of the models. Finally, the system produced outputs in the form of predictions or insights, which could be applied to decision making.

The architecture can be summarized as a sequence of stages:

1. Input data
2. Preprocessing
3. Exploratory data analysis
4. Model selection
5. Training and testing
6. Evaluation
7. Output and insights

4.8 Methodology

The methodology adopted during the internship was systematic and designed to cover the complete cycle of data analytics using Artificial Intelligence and Generative AI techniques. It ensured that the tasks were carried out in a logical order and that each stage contributed to the overall success of the project.

The process began with the identification of the problem statement and understanding the requirements of the project. Once the objectives were clear, datasets were collected and prepared for analysis. Data preprocessing was an important step, involving cleaning, normalization, and handling missing values to ensure that the data was reliable and suitable for further processing.

After preprocessing, exploratory data analysis was performed to study the characteristics of the dataset. This included identifying correlations, distributions, and trends that could guide the selection of appropriate models. Based on the analysis, suitable machine learning algorithms and generative AI models were chosen. These models were then implemented using Python and relevant libraries, trained on the prepared datasets, and tested to validate their performance.

The evaluation stage involved measuring the effectiveness of the models using metrics such as accuracy, precision, recall, and F1 score. The results were analyzed to identify strengths and limitations, and improvements were made where necessary. Finally, the outcomes were documented and presented in the form of insights and predictions that demonstrated the practical application of AI in data analytics.

4.9 Results and Discussion

The internship project produced results that demonstrated the practical application of Artificial Intelligence and Generative AI in data analytics. By following the structured methodology and system architecture, the tasks were completed successfully and provided valuable insights into how modern technologies can be used to solve real-world problems.

The first set of results came from data preprocessing and exploratory analysis. Cleaning and normalization improved the quality of datasets, while visualization revealed important patterns and correlations. This stage highlighted the importance of preparing data carefully before applying algorithms, as the accuracy of results depends heavily on the reliability of input data.

Machine learning models such as regression and classification were implemented using Scikit learn. These models achieved satisfactory levels of accuracy and provided predictions that aligned with expected outcomes. Generative AI models were also explored, and they proved useful in generating synthetic data that enhanced the scope of analysis. This showed how generative techniques can complement traditional analytics by expanding datasets and improving predictive performance.

Evaluation metrics such as accuracy, precision, recall, and F1 score were used to measure the effectiveness of the models. The results indicated that while traditional machine learning algorithms performed well on structured datasets, generative AI models offered additional flexibility and adaptability when dealing with complex or limited data.

The discussion of these results emphasizes the importance of combining different approaches in data analytics. The internship demonstrated that Artificial Intelligence and Generative AI are not only theoretical concepts but practical tools that can be applied to real problems. It also highlighted the need for continuous improvement, as model performance depends on careful tuning, proper data handling, and systematic evaluation.

Overall, the results confirmed that the objectives of the internship were achieved. The project provided hands-on experience, strengthened technical skills, and bridged the gap between academic learning and industry practice.

4.10 Conclusion

The internship on GenAI Powered Data Analytics provided a valuable opportunity to apply theoretical knowledge of Artificial Intelligence and Generative AI to practical scenarios. The structured program offered by Tata Forage allowed me to experience the complete cycle of data analytics, beginning from problem identification and dataset preparation to model implementation, evaluation, and generation of insights.

Through this internship, I gained hands-on experience with Python programming and essential libraries such as NumPy, Pandas, Scikit learn, TensorFlow, and Keras. I learned the importance of data preprocessing and exploratory analysis in ensuring the accuracy of results. The application of machine learning algorithms and generative AI models demonstrated how modern technologies can be used to solve complex problems and enhance predictive performance.

The project also helped in strengthening professional skills such as systematic working, documentation, and time management. More importantly, it bridged the gap between academic learning and industry practice, showing how classroom concepts can be transformed into real-world applications.

In conclusion, the internship was successful in achieving its objectives. It provided a holistic learning experience that combined technical knowledge with practical exposure. The skills and insights gained during this program will serve as a strong foundation for future projects and career opportunities in the field of Artificial Intelligence and Data Science.

4.11 Summary and Future Scope

The internship on GenAI Powered Data Analytics provided a comprehensive learning experience that combined theoretical knowledge with practical application. The project covered the complete cycle of data analytics, beginning with problem identification, dataset preparation, and exploratory analysis, followed by model implementation, evaluation, and generation of insights. Each stage was carried out systematically, ensuring that the objectives of the internship were achieved.

The summary of the work highlights the importance of data preprocessing, the role of machine learning algorithms, and the potential of generative AI in enhancing predictive analytics. The use of Python and its libraries created a strong technical foundation, while the structured methodology ensured that the tasks were aligned with industry practices. The results demonstrated that Artificial Intelligence and Generative AI are powerful tools for solving complex problems and can be applied effectively in real-world scenarios.

Looking towards the future, the scope of this project can be extended in several directions. Advanced generative models such as GANs and transformers can be explored to improve the quality of synthetic data and enhance predictions. Integration of big data platforms and cloud computing can expand the scale of analytics, allowing the handling of larger and more diverse datasets. Further research can also focus on improving model interpretability, ensuring that AI-generated insights are transparent and easily understood by decision makers.

Chapter 5: Conclusion and Recommendations

The internship on GenAI Powered Data Analytics provided a meaningful opportunity to apply theoretical concepts of Artificial Intelligence and Generative AI to practical scenarios. The structured program offered by Tata Forage ensured that the entire cycle of data analytics was covered, beginning with problem identification and dataset preparation, followed by model implementation, evaluation, and generation of insights.

The conclusion drawn from this project is that Artificial Intelligence and Generative AI are powerful tools that can transform raw data into actionable intelligence. The internship successfully achieved its objectives by strengthening technical skills in Python programming, machine learning, and generative modeling. It also emphasized the importance of data preprocessing, exploratory analysis, and systematic evaluation in ensuring reliable results. Beyond technical learning, the program helped in developing professional skills such as documentation, time management, and systematic working, which are essential for industry practice.

Based on the experience gained, several recommendations can be made for future work and improvement. Firstly, advanced generative models such as GANs and transformers should be explored to enhance the quality of synthetic data and improve predictive performance. Secondly, integration with big data platforms and cloud computing can expand the scale of analytics, allowing the handling of larger and more diverse datasets. Thirdly, greater focus should be placed on model interpretability, ensuring that AI-generated insights are transparent and easily understood by decision makers. Finally, continuous practice and research in the field of Artificial Intelligence and Data Science will help in keeping pace with rapid technological advancements.

In summary, the internship provided a holistic learning experience that bridged the gap between academic knowledge and industry application. The skills and insights gained during this program will serve as a strong foundation for future projects, academic research, and professional opportunities in Artificial Intelligence and Data Analytics

Certificate/Evidence

TATA

Forage

Inspiring and empowering future professionals

Syed Arshadul Haque

GenAI Powered Data Analytics Job Simulation

Certificate of Completion

August 21st, 2026

Over the period of August 2026, Syed Arshadul Haque has completed practical tasks in:

Exploratory data analysis and risk profiling
Predicting delinquency with AI
Business report and data storytelling for collections strategy
Implementing an AI-driven collections strategy

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a886d793a4eab45389d7181 | User Verification Code 6a85ce8bbe5acf73b1d8a44b | Issued by Forage

Forage / Syed Arshadul — congratulations on completing the Tata Data Visualisation: Empowering Business with Effective Insights Job Simulation

External Inbox ×

Forage <admin@theforage.com> Unsubscribe Wed, Aug 26, 11:15 PM (3 days ago)

to me

Hi Syed Arshadul,

We saw that you recently completed Tata's job simulation on Forage and the quality of your work stood out. It's clear you put a lot of effort into the simulation, so we wanted to say thank you and well done!

Our job is to support you as a standout participant on Forage on the journey to land a job or internship you'll love.

Take a moment to reflect, this is a massive achievement. By taking the time to work through the Tata job simulation, you've built the skills that Data

Bibliography/References:-

The following references were used to support the study and preparation of the Artificial Intelligence internship project. These resources provided useful information about Artificial Intelligence, Machine Learning, Python programming, data processing, algorithms, and practical implementation.

1. **Russell, S. J., &Norvig, P.** *Artificial Intelligence: A Modern Approach*. PearsonEducation.Thisbookprovidesfundamentalconceptsandprinciplesof Artificial Intelligence, including intelligent agents, problem-solving, learning, and AI applications.
2. **AurélienGéron.** *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media. This reference provides practical knowledge of MachineLearning,modeldevelopment,dataprocessing,andneuralnetworks.
3. WesMcKinney.*PythonforDataAnalysis*.O'ReillyMedia.Thisbookisuseful for understanding Python-based data manipulation, analysis, and preprocessing using libraries such as Pandas and NumPy.
4. Python Documentation. *Python Software Foundation*. The official documentationwasusefulforunderstandingPythonprogrammingconcepts, syntax, functions, and libraries.
5. Scikit-learn Documentation. The official Scikit-learn documentation provides informationaboutMachineLearningalgorithms,preprocessingmethods,model training, and evaluation techniques.
6. **TensorFlowDocumentation.**TheofficialTensorFlowdocumentationprovides guidance on developing and training Machine Learning and deep learning models.